

SHIELDDesk — SECURITY OPERATIONS PLATFORM

SSH Patch Orchestrator with LVM Rollback

Official Test Case Specification & Verification Matrix (v1.0.0)

TARGET ENGINE	UI INTEGRATION	TOTAL TEST CASES	CURRENT STATUS
Go Patch Microservice (Port 8004)	Security Scanner (/dashboard/scanner)	28 Test Cases Across 8 Modules	All Automated Suites Passed

Executive Quality Assurance Statement

This document defines the formal test specification for the ShieldDesk SSH Patch Orchestrator with LVM Snapshot Rollback. Every safety invariant specified in the architecture specification—including the zero-data-loss snapshot gate, deterministic state machine, strict SHA256 MITM rejection, semantic version validation, and deferred reboot recovery—is cataloged with explicit test steps, expected outcomes, and verification criteria.

Module 1: Pre-Check & Environment Validation (TC-PRE)

Verifies remote host diagnostics, operating system detection, active volume groups, free extent capacity, and baseline snapshot prerequisites.

TEST ID	TEST CASE TITLE	SEVERITY	EXECUTION STEPS & ASSERTION	RESULT
TC-PRE-01	Supported OS Detection (Ubuntu/Debian)	HIGH	Query <code>/etc/os-release</code> on target. Confirm engine identifies <code>apt</code> package manager and parses Ubuntu/Debian versions correctly.	PASSED AUTO
TC-PRE-02	Supported OS Detection (RHEL/Rocky/CentOS)	HIGH	Query <code>/etc/os-release</code> on RHEL target. Confirm engine identifies <code>dnf/yum</code> package manager adapters.	PASSED AUTO
TC-PRE-03	LVM Free Extents Headroom (>20%)	CRITICAL	Execute <code>vgs --units g</code> and assert volume group has at least 2 GB or >20% free space for the Copy-on-Write snapshot table.	PASSED AUTO
TC-PRE-04	Non-LVM Target Rejection	CRITICAL	Attempt to run orchestrator on host without LVM root partition. Assert transition to <code>PRECHECK_FAILED</code> with zero modifications applied.	PASSED AUTO
TC-PRE-05	Pre-Patch Baseline Version Capture	HIGH	Query <code>dpkg-query -W</code> or <code>rpm -q</code> prior to patch execution. Confirm baseline version string is recorded in audit log.	PASSED AUTO

Module 2: LVM Snapshot & Safety Gate (TC-SNP)

Verifies CoW snapshot generation, attribute flag validation, and structural enforcement of the Phase 4 Verification Gate.

TEST ID	TEST CASE TITLE	SEVERITY	EXECUTION STEPS & ASSERTION	RESULT
TC-SNP-01	Atomic LVM Snapshot Creation	CRITICAL	Run <code>lvcreate -s -L 2G -n snap_prepatch_<pkg> /dev/vg0/root</code> . Assert exit code 0 and logical volume presence.	PASSED AUTO
TC-SNP-02	LVM Attribute (lv_attr) Health Check	CRITICAL	Parse <code>lvs --noheadings -o lv_attr</code> . Confirm snapshot status is <code>swi-a-s---</code> (active, valid) and not <code>swi-I-s---</code> (invalid).	PASSED AUTO
TC-SNP-03	Strict Verification Gate Invariant	CRITICAL	Simulate snapshot creation error. Verify state machine structurally forbids transition to <code>PATCHING</code> ; stops at <code>SNAPSHOT_FAILED</code> .	PASSED AUTO
TC-SNP-04	Snapshot Naming Collision Prevention	MEDIUM	Ensure snapshot names include package name and unique timestamp/nonce to avoid collisions with prior snapshots.	PASSED AUTO

Module 3: Controlled Patch Execution (TC-PAT)

Verifies allowlisted structured operations, metacharacter sanitization, and network disconnection handling.

TEST ID	TEST CASE TITLE	SEVERITY	EXECUTION STEPS & ASSERTION	RESULT
TC-PAT-01	Allowlisted Vector Dispatch (apt-get)	HIGH	Execute <code>pkg.upgrade</code> on Ubuntu. Verify command array: <code>["apt-get", "install", "-y", "--only-upgrade", "pkg"]</code> .	PASSED AUTO
TC-PAT-02	Shell Injection Metacharacter Block	CRITICAL	Pass package name with injected characters: <code>openssl; rm -rf / or curl && reboot</code> . Assert validation rejects input before SSH execution.	PASSED AUTO
TC-PAT-03	Mid-Command SSH Connection Drop	CRITICAL	Simulate SSH socket closure during package install. Confirm outcome is flagged <code>DISCONNECTED</code> (never treated as success or normal fail).	PASSED AUTO

Module 4: Post-Patch Validation (TC-VAL)

Verifies version comparison routines, package query readbacks, and service reload confirmations.

TEST ID	TEST CASE TITLE	SEVERITY	EXECUTION STEPS & ASSERTION	RESULT
TC-VAL-01	Semantic Version Check (Ubuntu/Debian)	HIGH	Query <code>dpkg-query -W -f='\${Version}'</code> . Assert version comparison confirms installed version $\geq$ target version.	PASSED AUTO
TC-VAL-02	Version Mismatch Rollback Trigger	CRITICAL	Simulate package upgrade returning exit code 0 but installed version unchanged. Assert transition to <code>ROLLBACK_REQUIRED</code> .	PASSED AUTO
TC-VAL-03	Post-Patch Daemon Service Restart	MEDIUM	Specify restart service (e.g., <code>nginx</code>). Verify <code>systemctl restart nginx</code> executes and reports active state.	PASSED AUTO

Module 5: Rollback & Disaster Recovery (TC-ROL)

Verifies LVM snapshot merging, deferred reboot detection, reconnect polling, and escalation states.

TEST ID	TEST CASE TITLE	SEVERITY	EXECUTION STEPS & ASSERTION	RESULT
TC-ROL-01	Automatic Snapshot Merge (lvconvert)	CRITICAL	Trigger rollback. Verify execution of <code>lvconvert --merge /dev/vg0/snap_prepatch</code> and verify merge status.	PASSED AUTO
TC-ROL-02	Deferred Reboot Detection	CRITICAL	Detect stdout pattern "merging will occur on next activation". Verify orchestrator issues <code>reboot</code> and polls SSH for recovery.	PASSED AUTO
TC-ROL-03	Post-Rollback Baseline Verification	HIGH	After merge/reboot, query package version. Confirm version returned matches pre-patch baseline. Transition to <code>ROLLED_BACK_HUMAN_REVIEW</code> .	PASSED AUTO
TC-ROL-04	Catastrophic Escalation on Restore Failure	CRITICAL	Simulate failed snapshot merge or unreachable host after reboot. Confirm state reaches <code>ESCALATED_URGENT</code> and exits with code 2.	PASSED AUTO

Module 6: Security & Identity (TC-SEC)

Verifies strict SHA256 host key fingerprint validation, credentials handling, and audit redaction.

TEST ID	TEST CASE TITLE	SEVERITY	EXECUTION STEPS & ASSERTION	RESULT
TC-SEC-01	Strict SHA256 Host Key Verification	CRITICAL	Provide expected fingerprint matching host. Assert SSH connection completes successfully.	PASSED AUTO
TC-SEC-02	Mismatched Host Key Rejection (MITM)	CRITICAL	Provide forged fingerprint. Assert SSH client aborts handshake immediately before sending credentials.	PASSED AUTO
TC-SEC-03	Private Key & Token Redaction	HIGH	Inspect audit log output. Verify SSH private keys, passphrases, and bearer tokens are masked with [REDACTED] .	PASSED AUTO

Module 7: REST API & Microservice Integration (TC-API)

Verifies HTTP API endpoints on port 8004 and Next.js proxy integration.

TEST ID	TEST CASE TITLE	SEVERITY	EXECUTION STEPS & ASSERTION	RESULT
TC-API-01	Public Health Probe (GET /health)	HIGH	GET <code>http://localhost:8004/health</code> . Assert HTTP 200 and JSON with <code>status: "ok"</code> and version <code>"1.0.0"</code> .	PASSED AUTO
TC-API-02	Root Status Information (GET /)	MEDIUM	GET <code>http://localhost:8004/</code> . Assert HTTP 200 with service description, endpoints, and link to UI at port 3000.	PASSED AUTO
TC-API-03	Job Creation (POST /api/v1/jobs)	CRITICAL	Submit job JSON. Assert HTTP 201 with generated <code>job_id</code> and initial state <code>DETECTED</code> .	PASSED AUTO
TC-API-04	Live Log & State Polling (GET /api/v1/jobs/:id)	HIGH	Poll job endpoint during execution. Confirm logs array updates in real time with command progress.	PASSED AUTO
TC-API-05	Manual Rollback Trigger (POST /jobs/:id/rollback)	HIGH	Send rollback request. Assert orchestrator accepts request and initiates snapshot restore.	PASSED AUTO

Module 8: Web UI Console Verification (TC-UI)

Verifies the Next.js Security Scanner interface at `/dashboard/scanner#patch` .

TEST ID	TEST CASE TITLE	SEVERITY	EXECUTION STEPS & ASSERTION	RESULT
TC-UI-01	Real-Time Service Online Indicator	HIGH	Open Patch tab. Confirm banner updates to <code>Online (port 8004)</code> within 1 second via <code>/api/patch/health</code> .	PASSED UI
TC-UI-02	Form Validation & Safety Guards	HIGH	Leave required inputs blank and click Apply. Confirm red warning prevents submission.	PASSED UI
TC-UI-03	Live Terminal Log Streaming	HIGH	Submit patch job. Confirm terminal polls every 2s and renders color-coded command output.	PASSED UI
TC-UI-04	Emergency Rollback Action Button	HIGH	Ensure Rollback button is enabled during active job and disabled when idle.	PASSED UI